

Answer **all** the questions.

1. Alexandra, Betty, and Cassandra have a total of \$760. The ratio of Alexandra to Betty's money is 2 : 3, and the ratio of Betty to Cassandra's money is also 2 : 3. How much money does Alexandra have?

Answer \$ [2]

-
2. A company has a team bonding event, where each team consists of x male and y female employees. If there are 45 male and 27 female employees in total, what is the smallest value of $x + y$?

Answer [2]

3. The first four terms of a sequence, R_n , is as follows:

$$R_1 = \frac{1}{1} - \frac{1}{2} = \frac{1}{2}$$

$$R_2 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$R_3 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$R_4 = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$$

- (a) Write down R_{100} , in the same manner as above.

Answer $R_{100} = \dots\dots\dots$ [1]

- (b) Another sequence, S_n , is such that $S_n = R_1 + R_2 + \dots + R_n$.

Find the value of S_{100} .

Answer $S_{100} = \dots\dots\dots$ [2]

- (c) Describe what happens to the value of S_n as n increases to a very large value.

Answer $\dots\dots\dots$

$\dots\dots\dots$

$\dots\dots\dots$

[1]

4. A television is sold for \$1831.20 in Singapore, inclusive of a 9% Goods and Services Tax (GST).

(a) Find the price of the television before GST.

Answer \$ [2]

- (b) The same television is sold for AUD 1940 in Australia, where the exchange rate is AUD 1 = \$0.90 SGD. Is it cheaper to buy the television in Australia or Singapore? Explain your answer with mathematical calculations.

Answer The television is cheaper in because

.....

..... [2]

5. Tap P fills up a tank in 6 minutes while Tap Q fills up a tank in 10 minutes. How much time would it take to fill the tank if Taps P and Q were used to fill the tank together?

Answer minutes [2]

6. Write $\frac{2x+1}{3} - \frac{5x+3}{6}$ as a single fraction in the simplest form.

Answer [3]

7. Factorise each of the following expressions completely.

(a) $a^4 - 1$,

Answer [2]

(b) $4x^2 + 20xz - 2xy - 10yz$.

Answer [2]

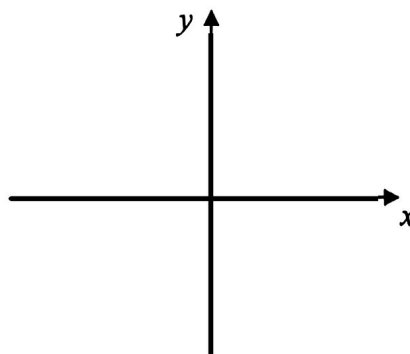
8. (a) Simplify $\frac{\sqrt[3]{8x^6}}{9x^{\frac{1}{2}}} \div \left(\frac{15}{5x}\right)^{-2}$, leaving your answer in positive index form.

Answer [2]

- (b) Hence, solve $\frac{\sqrt[3]{8x^6}}{9x^{\frac{1}{2}}} \div \left(\frac{15}{5x}\right)^{-2} = \frac{2}{5}$.

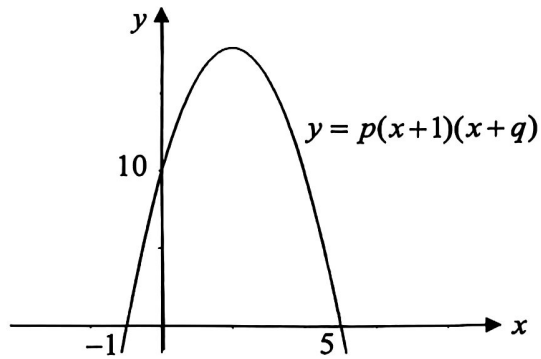
Answer $x =$ [2]

9. Sketch the graph of $y = 2^{x+1}$ on the axes provided, indicating any intercepts clearly on the graph.



Answer in space provided [2]

10. The diagram below shows the graph of the curve $y = p(x+1)(x+q)$.



- (a) Find the values of p and q .

Answer $p = \dots\dots\dots q = \dots\dots\dots$ [2]

- (b) Express $(x+1)(x+q)$ in terms of $(x+h)^2 + k$.

Answer $\dots\dots\dots$ [2]

- (c) State the equation of line of symmetry.

Answer $\dots\dots\dots$ [1]

- (d) Suggest what will happen to the graph if p is a positive value, but q remains unchanged.

Answer $\dots\dots\dots$

$\dots\dots\dots$ [2]

11. Kathy is buying cupcakes for 30 people. She needs to buy enough cupcakes so that every person can eat at least 2 cupcakes. Each cupcake cost \$2.90, and she is given a budget of \$210 to spend on the cupcakes.

By letting x be the number of cupcakes Kathy buys, form two inequalities in x and solve them.

Answer [3]

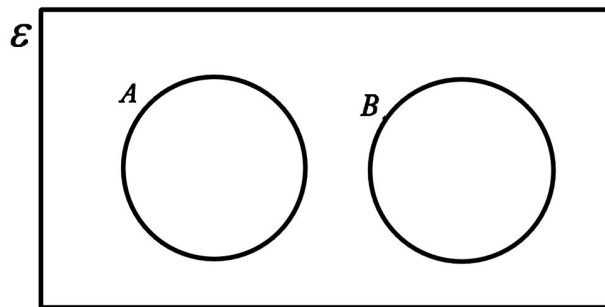
12. In a class test for 41 students, the mean mark was 18, and the median was 17. Suggest what might happen to the mean and median if a student who scored 21 marks is added to the class.

Answer

 [2]

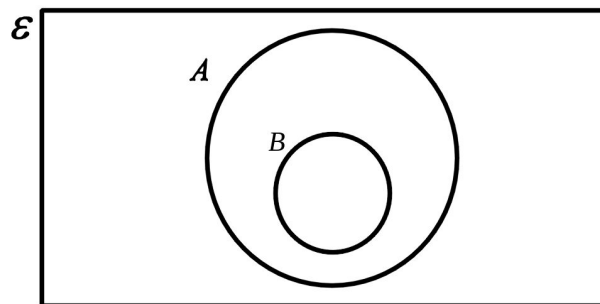
13. On each of the Venn diagrams below, shade the area representing $A \cap B'$.

(a) *Answer*



[1]

(b) *Answer*



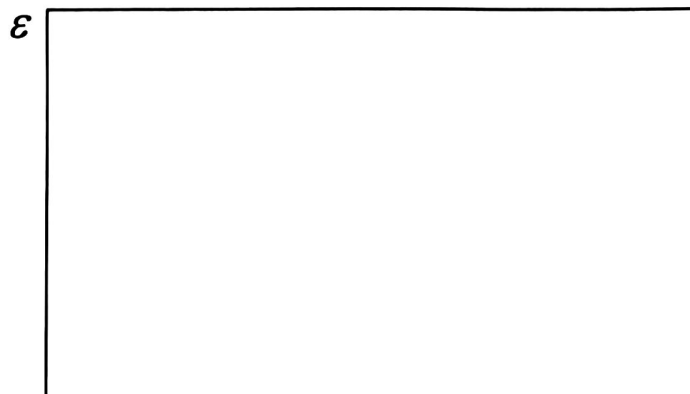
[1]

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14. In a class, each student learns one or more of the subjects, Chemistry, Physics, and Biology. All students who learn Physics also learn Chemistry. Some students who learn Biology also learn Chemistry, but no students learn both Biology and Physics.

Given that C represents the set of students learning Chemistry,
 P represents the set of students learning Physics, and
 B represents the set of students learning Biology,

draw sets C , P , and B in the Venn diagram below.

Answer



[2]

15. Laos is a country with a perimeter of 5270 km, and a land area of 236 800 km².
In the world map, the perimeter of Laos is 26.35 cm.

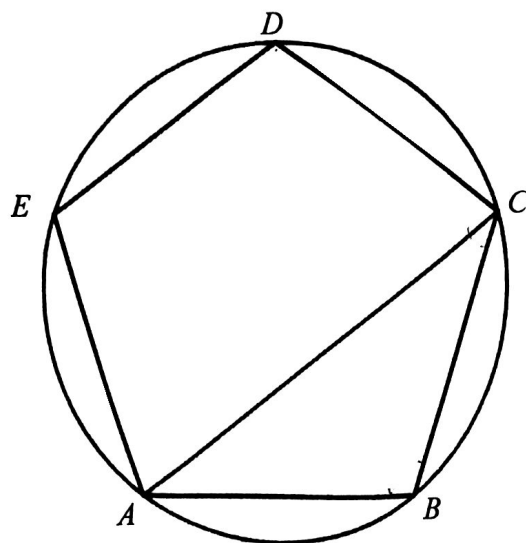
(a) The scale of the map is 1 : n . Find the value of n .

Answer n [2]

- (b) The land area of Kazakhstan is 11.5 times as large as Laos on the map.
Find the actual land area of Kazakhstan, leaving your answer in **standard form**, in km².

Answer km² [2]

16. A regular pentagon has all 5 vertices on the circumference of a circle.



- (a) Find angle ABC .

Answer $^{\circ}$ [1]

- (b) Find angle ACD .

Answer $^{\circ}$ [1]

- (c) Determine whether AC is the diameter of the circle. Explain your answer.

Answer

[2]

- (d) State the name of the shape $ACDE$ and explain your answer.

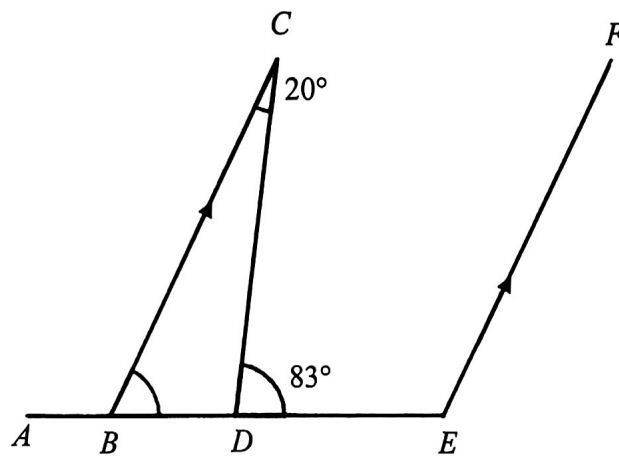
Answer $ACDE$ is a because.....

.....

.....

[2]

17. In the diagram below, $ABDE$ is a straight line. Angle $BCD = 20^\circ$, angle $CDE = 83^\circ$, and BC is parallel to EF .



By stating your reasons clearly, find

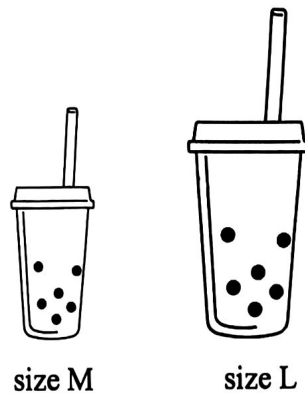
- (a) angle CBD ,

Answer $^\circ$ [1]

- (b) reflex angle DEF .

Answer $^\circ$ [2]

18. A bubble tea shop offers drinks in two geometrically similar cups, sizes M and L. The heights of the cups are 15 cm and 25 cm respectively. Each cup consists of tapioca pearls and tea, filled to the brim. Volume of tapioca pearls in both cups is the same.



- (a) Find the ratio of the volume of the cup M to cup L.

Answer [2]

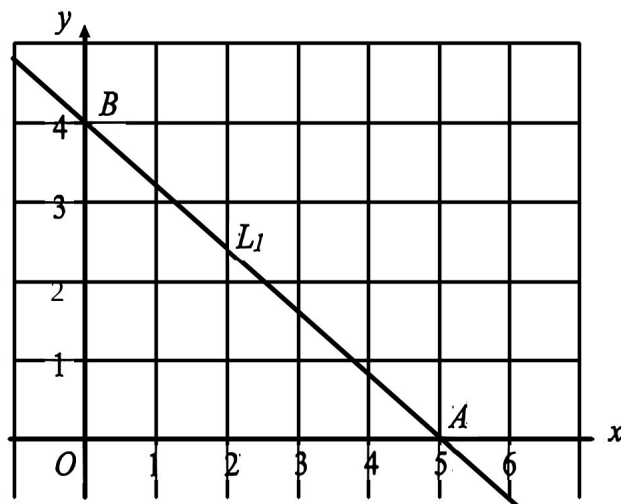
- (b) State the ratio of the base radius of the cup M to cup L.

Answer [1]

19. Given that $z = \frac{1}{7y}(x + 4y)^2$, express x in terms of y and z .

Answer $x =$ [2]

20. The diagram shows the line L_1 , cutting the x and y axes at points A and B respectively.



- (a) Another line, L_2 , is parallel to L_1 and passes through $(3, 0)$. Find the equation of line L_2 .

Answer [3]

- (b) Construct the bisector of angle OAB in the axes above.

Answer in the space above. [1]

- (c) The angle bisector in (b) cuts the y -axis at point C . Calculate the area of triangle ABC .

Answer units² [1]

21. The diagram below shows the speed-time graph of Ralph when he travels from home to a bookstore. He reaches the bookstore after 3 hours.



- (a) The area beneath the speed-time graph represents the distance travelled.
Find Ralph's average speed in km/h.

Answer km/h [3]

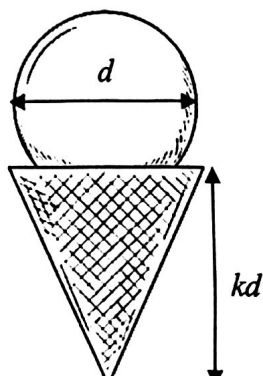
- (b) Ralph stops at the bookstore for 1 hour, then travels to the post office 10 km away at a constant speed of 20 km/h.

In the graph above, draw the speed-time graph of Ralph's journey from the bookstore to the post office.

Answer in the space above [2]

22. A cone of ice cream is formed using a spherical scoop of ice cream placed on top of a waffle cone. The diameter of the ice cream is equal to the diameter of the cone. The height of the cone is k times the diameter of the cone, where k is an integer.

Assume the waffle cone is of negligible thickness.



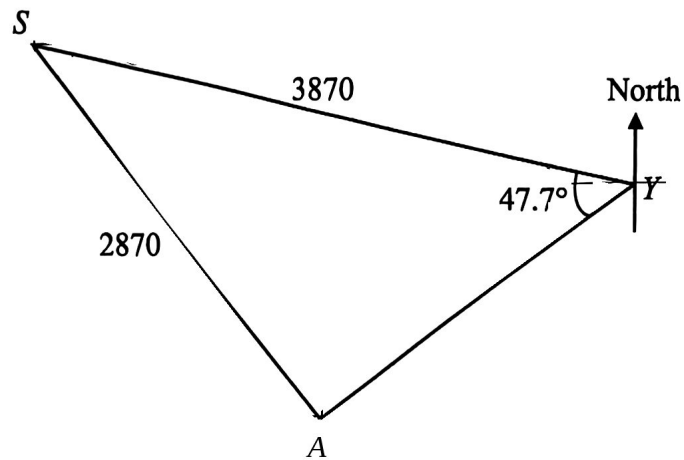
- (a) Given that the volume of the ice cream scoop is equal to the volume of the cone, find the value of k .

Answer $k = \dots\dots\dots$ [3]

- (b) If the radius of the cone is 2.1 cm, find the area of the waffle needed to make the cone.

Answer $\dots\dots\dots$ cm² [2]

23. Seattle (S), New York City (Y) and Austin (A) are three major cities in the United States of America. Seattle is 2870 km away from Austin, and 3870 km away from New York City. Angle SYA is 47.7° . The bearing of Seattle from New York City is 297° .



- (a) Find the bearing of New York City (Y) from Seattle (S).

Answer $^\circ$ [2]

- (b) Find the bearing of New York City (Y) from Austin (A).

Answer $^\circ$ [2]

- (c) Explain, with mathematical working, why there are two possible values for angle SAY .

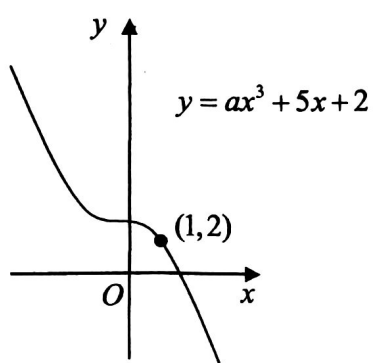
Answer

.....

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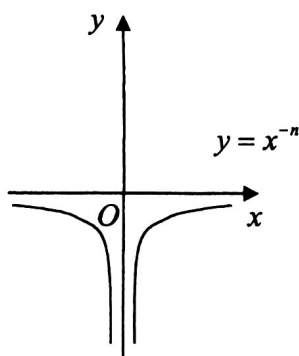
[3]

24. (a) Find the value of a .



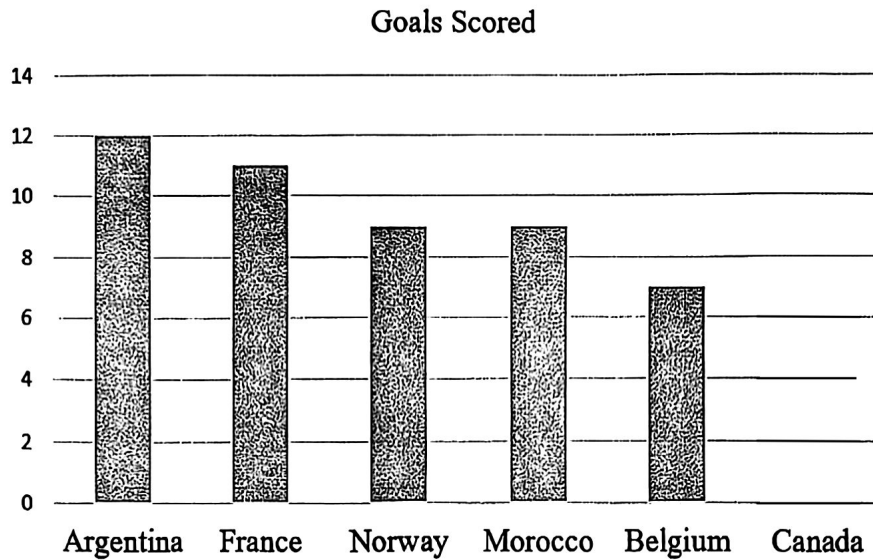
Answer $a =$ [1]

- (b) Find a possible value of n .



Answer $n =$ [1]

25. Below is a bar graph showing the number of goals scored by some countries in the 2026 Soccer World Cup. The number of goals scored by Canada is missing from the bar graph.



- (a) The actual mean of the number of goals scored by the 6 countries is 9. Complete the bar graph above by drawing the bar representing the number of goals scored by Canada.

Answer in the space above [2]

- (b) State the modal number of goals scored.

Answer [1]

- (c) Express the number of goals scored by the top performing country as a percentage of the total number of goals scored by all 6 countries.

Answer % [2]

- 1 (ai) Simplify $p^0 \div p^{-2}$, giving your answer in positive index notation.

Answer[1]

- (aii) Solve $\sqrt{4^x} \times 8^{x+1} = 0.5$.

Answer[3]

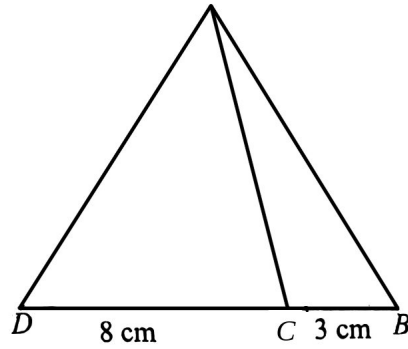
- (b) Solve the simultaneous inequalities $\frac{6x}{5} < \frac{x}{2} - 5 \leq 30 - 3x$.

Answer[2]

(c) Solve $\frac{5}{x} + \frac{3}{x+1} = \frac{7}{2x-1}$.

Answer $x = \dots\dots\dots$ or $\dots\dots\dots$ [4]

- 2 (a) In the diagram, BCD is a straight line. $AB = 7$ cm, $BC = 3$ cm, $AC = 6$ cm
 $DC = 8$ cm. Given that $\sin \hat{BAC} = \frac{1}{5}$ and without finding any angle, find
 the value of



- (i) $\sin \hat{ACB}$,

Answer[2]

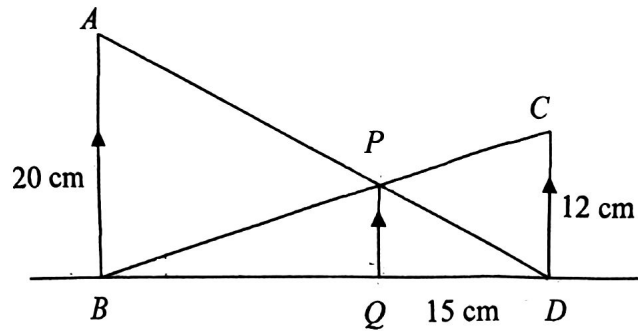
- (ii) $\sin \hat{ACD}$,

Answer[1]

- (iii) the area of triangle ADC .

Answercm² [2]

- 2 (b) The diagram below shows three vertical lines AB , PQ and CD . BQD is a horizontal line. $AB = 20$ cm, $CD = 12$ cm and $QD = 15$ cm.



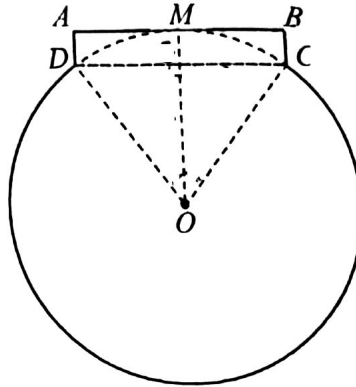
- (i) Write down a triangle that is similar to triangle BPQ .

Answer[1]

- (ii) Calculate the length of PQ .

Answercm [3]

- 3 The figure below is made up of a major segment of a circle with centre O spanned by arc CD , and a rectangle $ABCD$. AB is a tangent to the circle at M and $AB = 6$ cm and $BC = 1$ cm.



- (a) Show that the radius of the circle is 5 cm.

Answer shown in answer space [2]

- (b) Find angle DOC .

Answer.....° [2]

- (c) Find the area of the figure.

Answercm² [4]

- 4(a)** In a box, there are 12 bronze, 10 silver and 3 golden balls.
If three balls are drawn from the bag, one at a time, without replacement, find the probability that

- (i) only the third one is a golden ball,

Answer[1]

- (ii) two golden balls are drawn.

Answer[2]

- 4(b) Amy takes the driving test until she passes.
Each time Amy takes the test, the probability that she passes is 0.8.
Find the probability that Amy

(i) fails in the first three attempts

Answer[1]

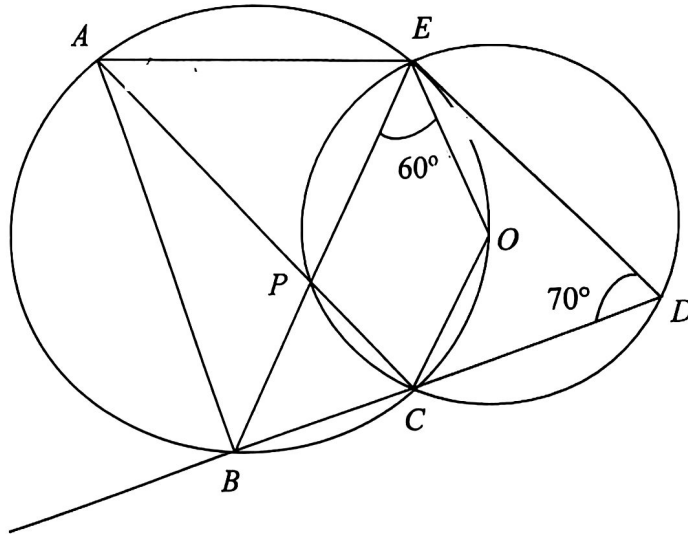
(ii) passes only at the second attempt,

Answer[1]

(iii) passes in one of the first three attempts.

Answer[2]

5



In the diagram, O is the centre of circle $CDEP$ and O is also on the circumference of the circle $ABCE$. $\hat{CDE} = 70^\circ$ and $\hat{OEP} = 60^\circ$. BPE , APC and BCD are straight lines.

(a) Find, stating your reasons clearly,

(i) $\angle COE$,

Answer $^\circ$ [1]

(ii) $\angle EAP$,

Answer $^\circ$ [1]

(iii) $\angle APB$,

Answer $^\circ$ [2]

(iv) $\angle OED$.

Answer $^{\circ}$ [2]

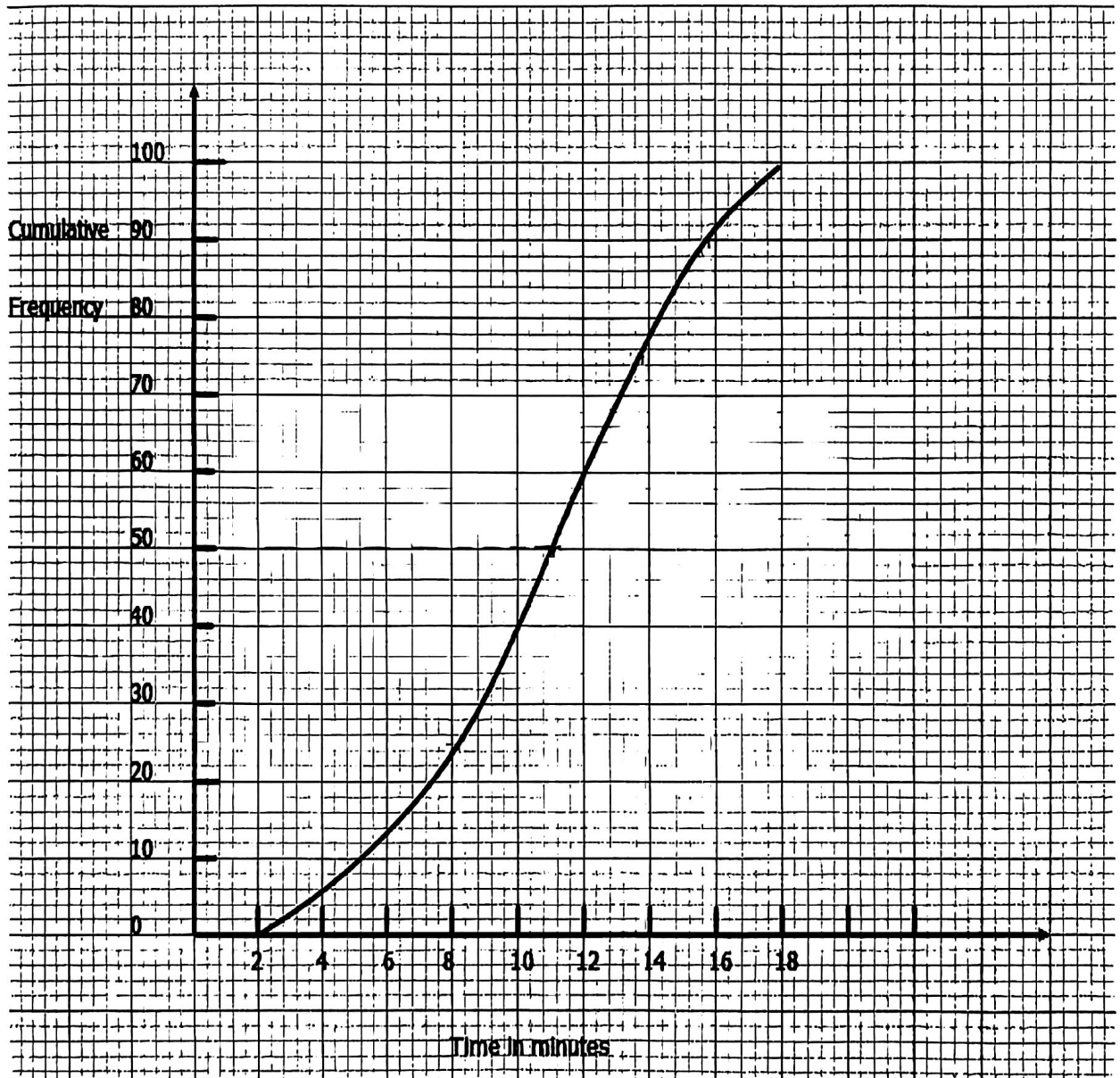
(b) Determine if AC is the diameter of the circle $ABCE$.

Answer shown in answer space [3]

- 6 (a) The time taken by 100 students to solve a mathematics problem successfully in a competition was recorded as follows:

Time (min)	$2 < t \leq 8$	$8 < t \leq 12$	$12 < t \leq 16$	$16 < t \leq 18$
No of students	24	36	x	y

A cumulative frequency curve of the above distribution was also drawn as shown below.



(i) Find the value of x and of y .

Answer $x = \dots\dots\dots, y = \dots\dots\dots$ [2]

(ii) Find the median.

Answer $\dots\dots\dots$ minutes [1]

(iii) Find the interquartile range.

Answer $\dots\dots\dots$ minutes [2]

(iv) If 10% of the students make it to the next stage of the competition, find the cut-off time for solving the problem before they can proceed to the next stage of the competition.

Answer $\dots\dots\dots$ minutes [1]

- (b) The waiting time, in minutes, for 50 customers at two bubble tea outlets are given as follows:

Outlet A					
Time (min)	$2 < t \leq 4$	$4 < t \leq 6$	$6 < t \leq 8$	$8 < t \leq 10$	$10 < t \leq 12$
Number of customers	3	25	10	10	2

Outlet B	
Mean = 8.2	
Standard Deviation = 2.63	

For Outlet A, calculate

- (i) the mean,

Answerminutes [1]

- (ii) the standard deviation.

Answerminutes [1]

- (iii) Compare briefly the waiting times of the two outlets in two different ways.

Answer _____

_____ [2]

- 7 The table below shows the number of t-shirts of different sizes sold in a day, together with their respective prices per piece, for 3 designs featuring the Singapore icons: Merlion, Changi Airport and MBS.

Design	Small per piece : \$10	Medium per piece : \$12	Large per piece \$15
Merlion	9	15	5
Changi Airport	10	16	3
MBS	12	18	6

- (i) Denoting the above matrix by P, evaluate $P \begin{pmatrix} 10 \\ 12 \\ 15 \end{pmatrix}$.

Answer[2]

- (ii) Explain what the numbers in your answer to part (i) represent.

Answer _____

 _____ [1]

- (iii) Evaluate $(1 \ 1 \ 1)P$.

Answer[1]

- (iv) What do the numbers in your answer to part (iii) represent?

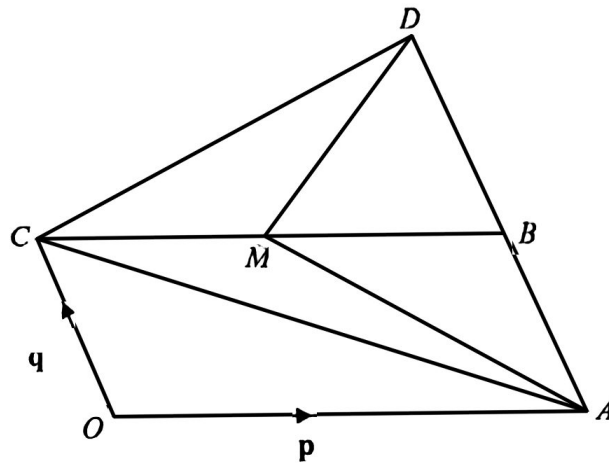
Answer _____

_____ [1]

- (v) Write down, **but do not evaluate**, a product of three matrices which give the total amount of money earned from the sales of all these t-shirts.

Answer [1]

8



In the diagram, $OABC$ is a parallelogram and M is the mid-point of BC . AB is produced to D such that $AB = BD$.

- (a) Given that $\overrightarrow{OA} = \mathbf{p}$ and $\overrightarrow{OC} = \mathbf{q}$, express the following vectors in terms of $\mathbf{p}$ and/or $\mathbf{q}$, giving each of your answers in the simplest form.

(i) $\overrightarrow{AC}$,

Answer[1]

(ii) $\overrightarrow{AM}$,

Answer[1]

(iii) $\overrightarrow{CD}$.

Answer[1]

- (b) Given that N is a point on CD such that $CD = 3CN$, show that $\overrightarrow{AN} = k \overrightarrow{AM}$ where k is a constant.

[2]

- (c) Find the numerical value of
- (i) $\frac{\text{area of triangle } CMD}{\text{area of triangle } CBD}$,

Answer[1]

- (ii) $\frac{\text{area of triangle } MND}{\text{area of triangle } CMD}$,

Answer[1]

9 Anne, Belle and Cindy each decided to buy a new car priced at \$85 000.

- (a) Anne paid for the car in cash and was given a discount. If she paid \$ 78 200 after the discount, calculate the percentage discount she received.

Answer% [2]

- (b) Belle decided to pay for her car using the hire purchase scheme. She paid a down payment of 15% of the price of the car, followed by 60 monthly instalments of \$1500. Calculate

- (i) the total cost of the car under this scheme,

Answer \$.....[2]

- (ii) the simple interest rate Belle had to pay.

Answer% [3]

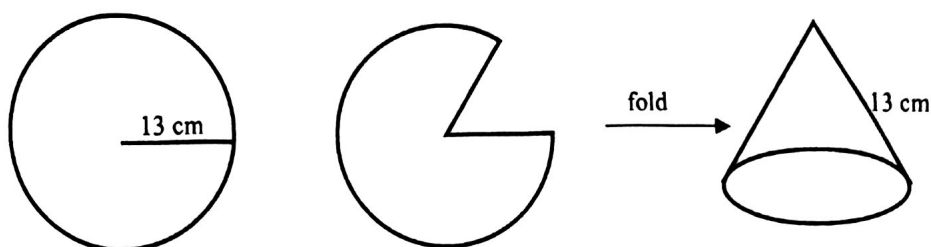
- (c) Cindy opted to buy the car by taking a loan of \$85 000. She would pay back the bank at the end of 4 years at an interest of 3.5% p.a. compounded monthly. Calculate the interest amount she had to pay.

Answer \$.....[3]

- (d) The value of the car depreciated by 10% at the end of each year, calculate the value of the car at the end of 3 years.

Answer \$ [2]

- 10** Amy sells chocolate from a stall. She is designing cone-shaped cups to hold them. She uses circular pieces of cardboard of radius 13 cm. The mass of 1 square metre of the cardboard is 220 grams.



She cuts a sector from a circular piece of cardboard, then joins the two straight edges of the remaining piece together to form an open cone. (The slant height of the cone is therefore equal to the radius of the original circle, 13 cm.)

By changing the size of the sector she removes, Amy can change the base radius and hence the volume of the cone.

- (a)** Amy makes a cone with base radius 5 cm. Given that the curved surface area of a cone is $\pi r l$ (where r is the base radius and l is the slant height), find the mass of this cone.

Answergrams [2]

(b) Amy cuts the sector so that the cone she forms has base radius x cm.

Show that the height of the cone is $h = \sqrt{169 - x^2}$ cm and the volume of the cone is $V = \frac{1}{3} \pi x^2 \sqrt{169 - x^2}$.

Answer

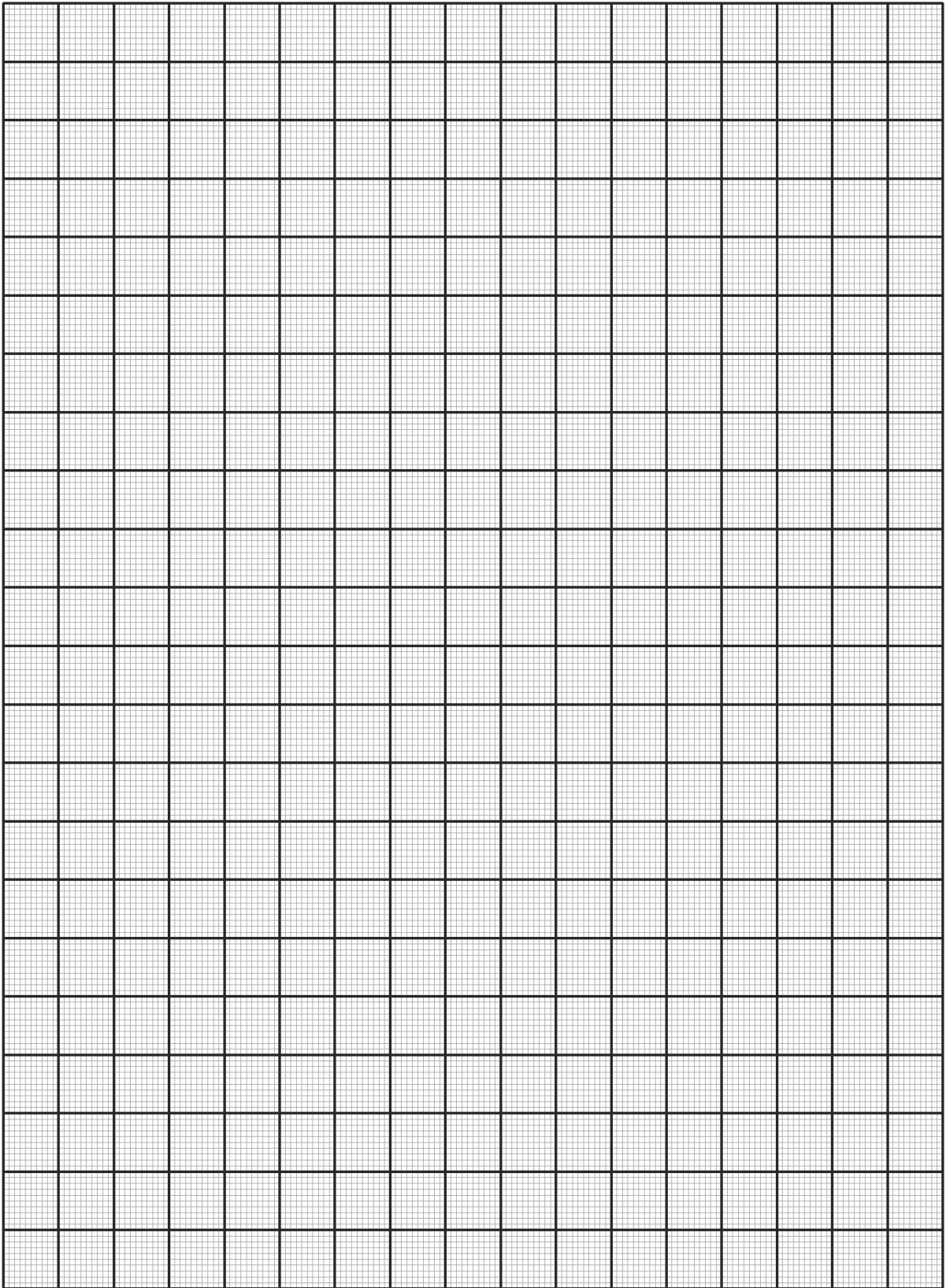
- (c) Amy wants to make a cone with volume of 800 cm^3 . She wants the mass of the cone to be as small as possible.

By drawing a suitable graph, find the value of x and work out the mass of the cone she will make.

Using the grid on page 24, plot your graph for $0 \leq x \leq 13$.

[7]

Answer



End of Paper